A Tightly Coupled Adaptive EKF for Indoor 3-D Positioning With Linear Acceleration Compensation

Lei Yao, *Member, IEEE*, Leehter Yao, *Senior Member, IEEE*, Shahaboddin Resalati, and Yeongwei Andy Wu

***Abstract*—This article presents a novel and robust 3-D localization approach that tightly integrates ultrawideband (UWB), inertial measurement unit (IMU), and magnetometer data to achieve high-precision attitude and 3-D position estimation. In contrast to existing methods, which often fail to account for the impacts of significant translational motion or are only effective under small linear accelerations (LAs), this work introduces an LA compensation method that dynamically adjusts measurement uncertainties in real time. This method addresses a critical limitation in previous approaches, enabling higher accuracy during significant motion. The proposed method leverages two extended Kalman filters (EKFs): six degrees of freedom (6-DoF) EKF for attitude estimation and nine degrees of freedom (9-DoF) EKF for 3-D position estimation, which are coupled for enhanced performance. The experimentally validated results show that the proposed method significantly outperforms state-of-the-art techniques, achieving root mean square errors of 1.96° for attitude estimation and 0.57 m for position estimation, while reducing computational complexity by 58.32%. This approach excels in challenging indoor environments with mixed line-of-sight (LOS) and non-line-of-sight (NLOS) conditions, demonstrating its robustness and applicability to real-world scenarios. This work presents a significant advancement in the field of 3-D localization, offering both theoretical contributions and practical improvements over existing systems.**

***Index Terms*—3-D positioning, adaptive, linear acceleration (LA) compensation, sensor fusion, tightly coupled extended Kalman filter (TC-EKF).**

I. Introduction

POSITIONING techniques are widely used in various fields. Outdoor applications include the global navigation satellite system (GNSS), which is used extensively in many fields, while approaches in indoor environments include radio-based positioning methods, such as Wi-Fi, Bluetooth, and ultrawideband (UWB), in addition to odometry and inertial positioning [1], [2]. In studies of indoor positioning methods, the algorithms used include least-squares, dead reckoning, and Kalman filter (KF) [3], [4], [5]. These methods can be classified into two groups: the first of which involves a system with a mobile node and several base stations. In this approach, the base stations are placed at fixed coordinate locations, and the positioning system's outputs describe the coordinates of the mobile node. The triangulation and least-squares algorithms are used to localize the mobile node by continuously measuring distances in terms of the time-of-flight (TOF) between the mobile node and its surrounding base stations. UWB is a TOF-based radio sensor operating in a wide frequency range of 3.1–10.6 GHz, with a channel bandwidth of 500 MHz. Since 2010, the wideband radio (more than 500 MHz) has been commercialized in the Qorvo DecaWave DWM1000 module [6]. By integrating UWB, precise positioning can be accomplished using several indoor positioning systems (IPSs), such as the Ubisense system, which can achieve an accuracy of 15 cm [7]. However, this level of accuracy is only possible under light-of-sight (LOS) conditions. Outliers, originating from radio blocking or multipath reflections, usually when radio signals propagate in mixed LOS and non-light-of-sight (NLOS) environments, can decrease the positioning accuracy significantly [8], [9], [10]. In contrast, the second group of methods relies on inertial navigation, where usually acceleration and angular rate are integrated over time to describe the device's position relative to an initial reference point. Dead reckoning [4] is an inertial-based approach, which provides a continuous series of positions. However, one notable issue is that the integration operation of inertial sensors will lead to cumulative error over time.

As seen, a single sensor has significant disadvantages; advances in multisensor fusion have led to substantial improvements. By using drift-free measurements, such as the TOF-based UWB ranging, real-time inclination angle computations, and drift-free heading estimations, the biases of accelerometers and gyroscopes can be successfully estimated, ensuring system convergence [11], [12], [13]. In general, sensor fusion strategies can be classified as either loosely coupled [23] or tightly coupled [30]. A loosely coupled approach typically fuses position-level outputs from different sensors. For example, UWB provides position estimates based on fixed anchors, and accelerometer data are double integrated to obtain position. The final position is determined by combining the independent position outputs. However, this approach does not explicitly model or correct the measurement errors of each sensor within the fusion framework. In contrast, a tightly coupled approach directly integrates raw measurements from

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sensors into a unified estimation process. In our case, UWB measurements are used not only to provide position information but also to correct accumulated drift and bias in the inertial sensors. This error-level integration enables more accurate and robust estimation. The tightly coupled multisensor fusion is a key theme in this article. Utilizing two extended KFs (EKFs) to combine data from inertial measurement unit (IMU), MAG, and UWB sensors [14], [15], [16]. The proposed method involves six degrees of freedom (6-DoF) EKF, nine degrees of freedom (9-DoF) EKF, and a tightly coupled framework integrating both EKFs. The fusion has two phases: 1) IMU and MAG fusion for attitude estimation using the 6-DoF EKF and 2) UWB and accelerometer fusion for 3-D positioning using the 9-DoF EKF. In addition, a tightly coupled framework is applied in this process, where the corrected attitude from the first phase is used in the second one, while the estimated accelerometer bias from the second phase is fed back into the first one. Notably, this framework demonstrates superior positioning accuracy compared with existing methods, along with a 58.32% reduction in computational complexity.

Although multisensor fusion methods have made good progress, especially in achieving high accuracy in bias estimation, there are still various types of errors, such as scale factor errors, misalignment of acceleration data, and uncertainty coming from linear acceleration (LA). These errors can be categorized into two types: 1) deterministic errors, such as bias, scale factor, and misalignment and 2) state-dependent errors that arise during motion. The deterministic errors have limited impact or their effects—such as scale factor and misalignment—can be mitigated through precalibration, while bias can be estimated using multisensor fusion approaches. In contrast, state-dependent errors are more dynamic and may lead to failures due to inaccurate a priori assumptions about the noise model. Therefore, the impact of those unexpected errors on accuracy remains a crucial challenge. Based on the existing literature, for example, Akbar et al. [17] and Kamal Mazha et al. [18] pointed out that the fusion of accelerometer and gyroscope fails to accurately estimate roll and pitch angles during translational motions due to an inaccurate noise model. In addition, numerous studies have reported that heading angle estimation based on gyroscope and magnetometer integration tends to be inaccurate when in proximity to artificial magnetic fields. This study does not explicitly evaluate magnetometer resilience, as magnetic field modeling is a complex topic with highly diverse approaches—many of which are environment-dependent and not generalizable across different indoor settings. Existing models often require environment-specific calibration, which limits their general applicability [19], [20], [21], [22]. Positioning based on the integration of acceleration and UWB/GNSS is also susceptible to obstacles and multiple reflections [23]. This uncertainty causes filters such as EKF to fail due to an inaccurate priori estimation of the noise model. Therefore, the adaptive adjustment of the Q matrix has been proposed in many studies to achieve better performance [17], [18], [24], [25], [26], [27].

This study focuses on state-dependent errors, specifically the uncertainty of LA in multi-sensor fusion. The LA term represents the acceleration due to motion excluding gravitational acceleration. Based on numerous studies, the accelerometer was found to provide drift-free inclination angle estimation, principally by computing the elements of gravity along the coordinate axes; however, the LA of the motion will introduce errors, which are difficult to identify and are commonly ignored [28], [29], [30], [31]. To improve the EKF's performance, this article introduces an LA compensation approach by adaptively adjusting the Q matrix of both EKFs. To the best of the authors' knowledge, the existing literature typically models the Q matrix as first-order low-pass-filtered white noise [24]. This ideal hypothesis works well when a vehicle is in a static state or undergoing uniform motion, i.e., scenarios in which the LA is equal to 0. However, experiments show that neglecting the LA can cause intense disturbances in positioning, especially during dramatic changes in motion. The proposed LA compensation method is also compared to two existing Q matrix adjustment methods, one of which applies a minimum volume covering ellipsoid (MVCE) method, while the other uses the covariance inflated (CI) solution. The MVCE utilizes a convex optimization strategy, in which the Mahalanobis distance between predicted and measured values is calculated using an elliptical normal distribution model. The observation function of MVCE can be adjusted based on the maximum Mahalanobis distance, which can further mitigate the inconsistencies between the predicted and measured values [19]. The CI method adjusts the Q matrix based on residual errors and can accordingly achieve better performance than that of the conventional KF [16]. Simulation and experimental results illustrate that the MCVE method can track medium acceleration motions, whereas the CI method can only track small acceleration motions. The reason is that the CI focuses on noise statistics, ensuring good performance under low LA conditions. In contrast, the MVCE constructs the Q matrix by adjusting the relative weighting between acceleration covariance data and the noise statistics. This process may amplify noise under low-LA conditions, while also reducing the ability to accurately reflect rapid acceleration changes. Both methods have limitations in terms of tracking full motion stages and require extensive parameter tuning, thus leading to implementation uncertainties. In contrast, the LA compensation method can achieve the most comprehensive and stable tracking performance among all the existing approaches. The main contributions of this work can be summarized as follows.

1) The uncertainty issue caused by accelerometer measurements due to the translational motion is addressed, and an LA compensation method is proposed.
2) The modified Q matrices in the proposed 6-DoF EKF and 9-DoF EKF designs achieve a near-optimal solution close to that of a 15-DoF EKF, especially under conditions of dramatic LA.
3) A tightly coupled framework integrating two EKFs is presented, where the proposed approach offers a 58.32% reduction in computing complexity.

The remainder of this article is organized as follows. Section II introduces the tightly coupled framework and the issues of uncertainty caused by dramatic shifts in motion.

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Section III provides a theoretical derivation of the Q matrix and explains the estimation process. Section IV analyses the computing complexity. Sections V and VI present a comparison of the results with those of existing methods. Finally, Section VII presents the conclusion of this article.

II. Problem Formulation

In IPS, both the measured angular rate vector $\boldsymbol{\omega}_m$ from the gyro and the acceleration vector $\boldsymbol{a}_m$ from the accelerometer contain bias $\boldsymbol{b}$ and noise $\boldsymbol{n}$. Typically, the bias is a nonzero constant, while the noise can be modeled as a random process following a Gaussian distribution, in which the distribution of noise is given by $N(0, \sigma_g^2)$, where $N(\mu, \sigma^2)$ denotes a Gaussian probability density function with mean μ and variance σ^2. Given the above, $\boldsymbol{\omega}$ and $\boldsymbol{a}$ can be written as follows:

$$\boldsymbol{\omega} = \boldsymbol{\omega}_m - \boldsymbol{b}_g - \boldsymbol{n}_g \tag{1}$$

$$\boldsymbol{a} = \boldsymbol{a}_m - \boldsymbol{b}_a - \boldsymbol{n}_a \tag{2}$$

where the subscript terms g and a represent the gyro and accelerometer, respectively.

The direction cosine matrix (DCM), which is used to convert coordinates between different systems or states, is denoted by $\boldsymbol{C}$ here. For example, $\boldsymbol{C}_B^N$ represents the DCM for conversions from the body frame (B-frame) to the navigation frame (N-frame), while the $\boldsymbol{C}(\delta\boldsymbol{\psi})$ term in (3) represents the angle error DCM used to correct the estimated $\hat{\boldsymbol{C}}_B^N$ to obtain the real $\boldsymbol{C}_B^N$. Equations (3)–(6) illustrate the relationship between the uncertainty angle error and the DCM

$$\boldsymbol{C}_B^N = \boldsymbol{C}(\delta\boldsymbol{\psi})\hat{\boldsymbol{C}}_B^N \tag{3}$$

$$\boldsymbol{C}(\delta\boldsymbol{\psi}) = \boldsymbol{I}_{3\times 3} + [\delta\boldsymbol{\psi}\times] \tag{4}$$

$$\delta\boldsymbol{\psi} = \delta\boldsymbol{\psi}_G + \boldsymbol{\varepsilon}_\psi \tag{5}$$

$$\hat{\boldsymbol{C}} = \left(q_0^2 - \boldsymbol{q}^T\boldsymbol{q}\right)\boldsymbol{I}_{3\times 3} - 2q_0[\boldsymbol{q}\times] + 2\boldsymbol{q}\boldsymbol{q}^T. \tag{6}$$

In the above equations, the estimated $\hat{\boldsymbol{C}}$ propagation can be computed from the quaternion using (6), where q_0 and $\boldsymbol{q}$ are the real and imaginary parts of the quaternion, respectively, $\boldsymbol{I}_{3\times 3}$ is a 3×3 identity matrix, and $[\cdot\times]$ is the skew-symmetric operator. The vector $\boldsymbol{q} = [\omega_x, \omega_y, \omega_z]^T$ can be computed form the angular rate ω. Considering the uncertainty from the LA, the attitude error $\delta\boldsymbol{\psi}$ should be represented as an attitude error $\delta\boldsymbol{\psi}_G$ from ground noise to which the uncertainty angle error $\boldsymbol{\varepsilon}_\psi$ from LA is added, as shown in (5). The attitude error is then further used in (4) to finally yield (3).

Another problem in this field relates to the poor performance of existing methods in terms of both accuracy and efficiency. Specifically, 15-DoF EKFs have been used in algorithms such as SHFAF [14] and CKF [32]. Due to the uncertainty term $\boldsymbol{\varepsilon}_\psi$ noted in (5), it is difficult for the 15-DoF EKF to obtain optimized solutions encompassing various movements, because these methods integrate many states (total of 15 states forming a 15×15 matrix), making them extremely computationally complex. In addition, these methods focus more on position than attitude and tend to ignore the uncertainty issue, leading to decreased accuracy. To improve this aspect, this article presents a tightly coupled framework in which the estimation procedure is divided into two phases. In the first phase, we design a 6-DoF EKF for attitude estimation, and in the second phase, we propose a 9-DoF EKF for localization. Benefiting from this framework, the uncertainty $\boldsymbol{\varepsilon}_\psi$ can be theoretically derived, as detailed in Section III, and the system's positioning accuracy improves markedly, along with a 58.32% reduction in computational complexity.

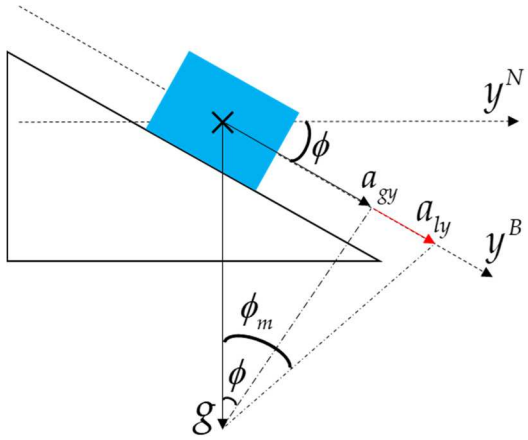


Fig. 1. Pitch angle measurement error due to LA.

III. Proposed Multisensor Fusion Algorithms

The 15-DoF EKF defines the error states as $\delta\boldsymbol{x} = [\delta\boldsymbol{p}, \delta\boldsymbol{v}, \delta\boldsymbol{\psi}, \delta\boldsymbol{b}_a, \delta\boldsymbol{b}_g]^T$, where $\delta\boldsymbol{p}, \delta\boldsymbol{v}, \delta\boldsymbol{\psi}, \delta\boldsymbol{b}_a$, and $\delta\boldsymbol{b}_g$ are the triaxis position error, velocity error, attitude error, accelerometer bias error, and gyro-bias error, respectively. In theory, a standard 15-DoF EKF may fail to correct angle errors during transitional motion. As shown in Fig. 1, the accelerometer measurements used in the 15-DoF EKF's correction process contain both the projection of gravitational acceleration a_{gy} and the LA a_{ly}. For example, if the object has an estimated pitch angle ϕ, the actual measured pitch would be $\phi_m = \arcsin((a_{gy} + a_{ly})/g)$, which includes the LA components, whereas the ideal pitch measurement—reflecting only the gravitational components—is $\phi = \arcsin(a_{gy}/g)$, where g represents the gravitational acceleration. Therefore, the LA components lead to significant angular estimation errors in the 15-DoF EKF. To address this issue, our proposed method divides the error states into a $\delta\boldsymbol{\gamma} = [\delta\boldsymbol{\psi}_G, \delta\boldsymbol{b}_g]^T$ term for a 6-DoF EKF and a $\delta\boldsymbol{\beta} = [\delta\boldsymbol{p}, \delta\boldsymbol{v}, \delta\boldsymbol{b}_a]^T$ term for a 9-DoF EKF, thus providing a clear framework for how to mitigate this issue in Sections III-A–III-D. The LA based adjustment method is hereafter referred to as ALA.

A. ALA-Based Error Covariance Estimator

The uncertainty issue from LA is derived in this section. The uncertainty angle error vector $\boldsymbol{\varepsilon}_\psi$ can be defined as $[\varepsilon_\phi, \varepsilon_\theta, \varepsilon_\psi]^T$, where ϕ, θ, and ψ denote the roll, pitch, and yaw with respect to the X-, Y-, and Z-axes, respectively. From the LA error model described in [24], the measured roll and pitch errors can be calculated as follows:

$$\begin{cases} \varepsilon_\phi = \arctan\left(\hat{a}_{ly}^B / (g\cos\ \phi)\right) \\ \varepsilon_\theta = \arctan\left(\hat{a}_{lx}^B / (g\cos\ \theta)\right) \end{cases} \tag{7}$$

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where g is the gravitational constant. The measured LA vector, $\hat{\boldsymbol{a}}_l^B = (\hat{a}_{lx}^B, \hat{a}_{ly}^B)^T$, can be estimated by the derivative of velocity, i.e., $\hat{\boldsymbol{a}}_l^B = \hat{\boldsymbol{C}}_N^B \dot{\boldsymbol{v}}^N$. Beginning with the first quadrant of the coordinates, the extrema of arc tan x as x ranges in $[0, \infty)$ will be $\lim_{x\to 0} \arctan x = x$ and $\lim_{x\to\infty} \arctan\ x = \pi/2$; thus, the extrema of (7) are given by

$$\begin{cases} \lim\limits_{\varepsilon_\phi \to 0}\ \varepsilon_\phi = \hat{a}_{ly}^B \big/ (g \cos\ \phi) < \pi/2 \\ \lim\limits_{\varepsilon_\theta \to 0}\ \varepsilon_\theta = \hat{a}_{lx}^B \big/ (g \cos\ \theta) < \pi/2. \end{cases} \tag{8}$$

To simplify designs, we consider a small value of $\hat{\boldsymbol{a}}_l^B$ and constant angles of ϕ and θ as vehicles tend to remain largely stable in real applications. Therefore, the mean of $\hat{\boldsymbol{a}}_l^B$ over a given sampling period can be considered 0, making its variance proportional to the magnitude of LA. The variance can be defined as $\lambda \left(\hat{a}_l^B\right)^2$, where λ can be experimentally determined or computed in real time. To obtain a more general expression accounting for the possibility that the angle may fall into any quadrant, we calculate the covariance of $\boldsymbol{\varepsilon}_\psi$ and combine the extrema range given in (8), thus yielding

$$\begin{cases} \operatorname{diag}\left(\lambda \left(\hat{a}_{ly}^B / g \cos\ \phi\right)^2,\quad \lambda \left(\hat{a}_{lx}^B \big/ g \cos\ \theta\right)^2, 0\right), \\ \qquad\qquad\qquad\qquad \text{if } \sum_{11}, \sum_{22} < \pi^2/4 \\ \operatorname{diag}\left(\pi^2/4, \pi^2/4, 0\right), \quad \text{if } \sum_{11}, \sum_{22} \geq \pi^2/4 \end{cases} \tag{9}$$

where $\sum_{11} = \lambda(\hat{a}_{ly}^B / g \cos\ \phi)^2$ and $\sum_{22} = \lambda(\hat{a}_{lx}^B / g \cos\ \theta)^2$. As we know, the measured LA, $\hat{a}_l^B$, also contains noise, which can be modeled as a random process that follows a Gaussian distribution with a mean of 0 and a variance of σ_a^2. Thus, the variance of the true LA with this noise removed would be $|\lambda(\hat{a}_l^B)^2 - \sigma_a^2|$. Hence, (9) can be rewritten as follows:

$$\begin{cases} \operatorname{diag}\left(\left|\lambda \left(\hat{a}_{ly}^B\right)^2 - \sigma_a^2\right| \Big/ (g \cos\ \phi)^2, \right. \\ \qquad \left. \left|\lambda \left(\hat{a}_{lx}^B\right)^2 - \sigma_a^2\right| \Big/ (g \cos\ \theta)^2, 0\right), \quad \text{if } \sum_{11}, \sum_{22} < \pi^2/4 \\ \operatorname{diag}\left(\pi^2/4, \pi^2/4, 0\right), \qquad\qquad \text{if } \sum_{11}, \sum_{22} \geq \pi^2/4. \end{cases} \tag{10}$$

Note that (10) is the proposed ALA-based Q matrix estimator $\sum_{\text{ALA}}$.

B. 6-DoF ALA-EKF for Attitude Estimation

The 6-DoF EKF integrates IMU and MAG sensors to estimate attitude and gyro bias, as detailed below. The 6-DoF vector $\delta\boldsymbol{\gamma}$ and its propagation are expressed in (11) and (12), and (13) is an extension of the KF for nonlinear systems obtained by approximating the state transition matrix $\boldsymbol{\Phi}$ through a first-order derivative. From study [33], the 6-DoF EKF's F and H matrices can be obtained through (14)–(17)

$$\delta\boldsymbol{\gamma} = \left[\delta\boldsymbol{\psi}_G, \delta\boldsymbol{b}_g\right]^T \tag{11}$$

$$\delta\boldsymbol{\gamma}_{t|t-1} = \boldsymbol{\Phi}_{\gamma,t-1}\delta\boldsymbol{\gamma}_{t-1|t-1} \tag{12}$$

$$\boldsymbol{\Phi}_{\gamma,t-1} = \exp\left[\boldsymbol{F}_{\gamma,t-1}\Delta t\right] \approx \boldsymbol{I}_{6\times 6} + \boldsymbol{F}_{\gamma,t-1}\Delta t \tag{13}$$

$$\delta\dot{\boldsymbol{\psi}}_G = -\left[\hat{\boldsymbol{C}}_B^N \boldsymbol{\omega}^B \times\right] \delta\boldsymbol{\psi} - \hat{\boldsymbol{C}}_B^N \delta\boldsymbol{b}_g - \hat{\boldsymbol{C}}_B^N \boldsymbol{n}_g \tag{14}$$

$$\boldsymbol{F}_\gamma = \begin{bmatrix} -\left[\hat{\boldsymbol{C}}_B^N \boldsymbol{\omega}^B \times\right] & -\hat{\boldsymbol{C}}_B^N \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{0}_{3\times 3} \end{bmatrix} \tag{15}$$

$$\boldsymbol{z}_\gamma = \left[\boldsymbol{a}_m^B - \hat{\boldsymbol{C}}_N^B \boldsymbol{a}_g^N, \boldsymbol{m}_m^B - \hat{\boldsymbol{C}}_N^B \boldsymbol{m}^N\right]^T = \boldsymbol{H}_\gamma \delta\boldsymbol{\gamma} + \boldsymbol{n}_\gamma \tag{16}$$

$$\boldsymbol{H}_\gamma = \begin{bmatrix} \hat{\boldsymbol{C}}_N^B \left[\boldsymbol{a}_g^N \times\right] & \boldsymbol{0}_{3\times 3} \\ \hat{\boldsymbol{C}}_N^B \left[\boldsymbol{m}^N \times\right] & \boldsymbol{0}_{3\times 3} \end{bmatrix} \tag{17}$$

where $\boldsymbol{m}$ is the strength of the magnetic field. Assuming that $\boldsymbol{n}_\gamma$ in (16) follows $N\left(0, \sum_1\right)$, we obtain:

$$\sum_1 = \operatorname{diag}\left(\sigma_a^2 \boldsymbol{I}_{3\times 3}, \sigma_m^2 \boldsymbol{I}_{3\times 3}\right) \tag{18}$$

where the subscript a and m terms represent the accelerometer and magnetometer, respectively. By accounting for the proposed uncertainty error $\boldsymbol{\varepsilon}_\psi$ described in (5), expressions (4), (14), and (16) can be rewritten as follows:

$$\boldsymbol{C}(\delta\boldsymbol{\psi}) = \boldsymbol{I}_{3\times 3} + \left[\left(\delta\boldsymbol{\psi}_G + \boldsymbol{\varepsilon}_\psi\right) \times\right] \tag{19}$$

$$\delta\dot{\boldsymbol{\psi}}_G = -\left[\hat{\boldsymbol{C}}_B^N \boldsymbol{\omega}^B \times\right] \left(\delta\boldsymbol{\psi}_G + \boldsymbol{\varepsilon}_\psi\right) - \hat{\boldsymbol{C}}_B^N \delta\boldsymbol{b}_g - \hat{\boldsymbol{C}}_B^N \boldsymbol{n}_g \tag{20}$$

$$\begin{aligned} \boldsymbol{z}'_\gamma &= \left[\boldsymbol{a}_m^B - \hat{\boldsymbol{C}}_N^B \hat{\boldsymbol{a}}_l^N - \hat{\boldsymbol{C}}_N^B \boldsymbol{a}_g^N, \boldsymbol{m}_m^B - \hat{\boldsymbol{C}}_N^B \boldsymbol{m}^N\right]^T \\ &= \boldsymbol{H}_\gamma \delta\boldsymbol{\gamma} + \boldsymbol{n}_\gamma + \begin{bmatrix} \hat{\boldsymbol{C}}_N^B \left[\boldsymbol{a}_g^N \times\right] \boldsymbol{\varepsilon}_\psi \\ \hat{\boldsymbol{C}}_N^B \left[\boldsymbol{m}^N \times\right] \boldsymbol{\varepsilon}_\psi \end{bmatrix} + \begin{bmatrix} -\hat{\boldsymbol{C}}_N^B \hat{\boldsymbol{a}}_l^N \\ \boldsymbol{0}_{3\times 1} \end{bmatrix} \end{aligned} \tag{21}$$

respectively. Therefore, the novel $\boldsymbol{Q}$ and $\boldsymbol{R}$ matrices of the proposed 6-DoF ALA-EKF, which are derived from (20) and (21), respectively, can be represented as follows:

$$\boldsymbol{Q}_\gamma = \sum_g + \sum_\gamma \tag{22}$$

$$\boldsymbol{R}_\gamma = \sum_1 + \sum_2 \tag{23}$$

where $\sum_g$ is the covariance matrix of the gyro noise, denoted by $\sum_g = \operatorname{diag}(\sigma_g^2 \boldsymbol{I}_{3\times 3}, \sigma_{\text{grw}}^2 \boldsymbol{I}_{3\times 3}), \sigma_g^2$ is the variance of the gyroscope, and σ_{grw}^2 is the variance of the random walk noise of the gyro biases. $\sum_\gamma$ is the error covariance matrix of the uncertainty error, denoted by $\sum_\gamma = \operatorname{diag}\left([\hat{\boldsymbol{C}}_B^N \boldsymbol{\omega}^B \times] \sum_{\text{ALA}} [\hat{\boldsymbol{C}}_B^N \boldsymbol{\omega}^B \times]^T, \boldsymbol{0}_{3\times 3}\right)$, where $\sum_{\text{ALA}}$ is described in (10), and $\sum_1$ is defined in (18). Let $\sum_2 = \operatorname{diag}\left(\sum_{d11}, \sum_{d22}\right)$, where $\sum_{d22} = \hat{\boldsymbol{C}}_N^B [\boldsymbol{m}^N \times] \sum_{\text{ALA}} [\boldsymbol{m}^N \times]^T \hat{\boldsymbol{C}}_B^N$ and $\sum_{d11} = \hat{\boldsymbol{C}}_N^B [\boldsymbol{a}_g^N \times] \sum_{\text{ALA}} [\boldsymbol{a}_g^N \times]^T \hat{\boldsymbol{C}}_B^N + \operatorname{diag}(\lambda^N \hat{a}_{lx}^2, \lambda^N \hat{a}_{ly}^2, 0)$, and the LA vector can be calculated as the differential of the velocity, i.e., $\hat{\boldsymbol{a}}_l^N = \dot{\boldsymbol{v}}^N$.

C. 9-DoF ALA-EKF for 3-D Position Estimation

The 9-DoF EKF estimates 3-D position by fusing data from UWB and accelerometer. For localization in the mixed LOS and NLOS environments, the actual distance of UWB can be expressed as follows:

$$\begin{aligned} d_i = d_{i,m} - n_\beta = \big((x_i - x_m)^2 + (y_i - y_m)^2 \\ + (z_i - z_m)^2\big)^{1/2} - n_\beta \end{aligned} \tag{24}$$

where the ranging noise n_β distributed by a two-mode Gaussian mixture model can be represented as $(1-\tau)N(0, \sigma_1^2) +$

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$\tau N(\mu_2, \sigma_2^2)$. The noise components for LOS and NLOS conditions are $N(0, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively, while the probabilities of LOS and NLOS are $1-\tau$ and τ ($0 \le \tau \le 1$), respectively. The vector $[x_i, y_i, z_i]^T$ denotes the known coordinates of the UWB base stations, where $i = 1, 2, 3, \ldots, N$ (where, N denotes the number of base stations), and $[x_m, y_m, z_m]^T$ denotes the unknown source coordinates of the UWB mobile node. Furthermore, the $d_{i,m}$ term with a subscript m represents the measured distance between the UWB mobile node and the ith base station, while d_i is the true distance.

The estimation steps of 9-DoF EKF are similar to those of the 6-DoF EKF. The 9-DoF vector $\delta\boldsymbol{\beta}$ and its prediction equations can be written as follows:

$$\delta\boldsymbol{\beta} = \left[\delta\boldsymbol{p}, \delta\boldsymbol{v}, \delta\boldsymbol{b}_a\right]^T \tag{25}$$

$$\boldsymbol{\beta}_{t|t-1} = \boldsymbol{\Phi}_{\beta,t-1}\boldsymbol{\beta}_{t-1|t-1} \tag{26}$$

$$\boldsymbol{\Phi}_{\beta,t-1} = \exp\left[\boldsymbol{F}_{\beta,t-1}\Delta t\right] \approx \boldsymbol{I}_{9\times9} + \boldsymbol{F}_{\beta,t-1}\Delta t \tag{27}$$

$$\delta\dot{\boldsymbol{v}} = -\left[\hat{\boldsymbol{C}}_B^N\boldsymbol{a}^B\times\right]\delta\boldsymbol{\psi} - \hat{\boldsymbol{C}}_B^N\delta\boldsymbol{b}_a - \hat{\boldsymbol{C}}_B^N\boldsymbol{n}_a \tag{28}$$

$$\boldsymbol{F}_\beta = \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & -\hat{\boldsymbol{C}}_B^N \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix}. \tag{29}$$

Based on the proposed uncertainty error $\boldsymbol{\varepsilon}_\psi$ in (5), (28) can be rewritten as follows:

$$\delta\dot{\boldsymbol{v}} = -\left[\hat{\boldsymbol{C}}_B^N\boldsymbol{a}^B\times\right]\left(\delta\boldsymbol{\psi}_G + \boldsymbol{\varepsilon}_\psi\right) - \hat{\boldsymbol{C}}_B^N\delta\boldsymbol{b}_a - \hat{\boldsymbol{C}}_B^N\boldsymbol{n}_a. \tag{30}$$

The novel $\boldsymbol{Q}$ matrix for the proposed 9-DoF ALA-EKF can be derived as follows:

$$\boldsymbol{Q}_\beta = \sum_a + \sum_\beta \tag{31}$$

where

$$\sum_a = \mathrm{diag}\left(\boldsymbol{0}_{3\times3}, \sigma_a^2\boldsymbol{I}_{3\times3}, \sigma_{\mathrm{arw}}^2\boldsymbol{I}_{3\times3}\right)$$

$$\sum_\beta = \mathrm{diag}\left(\boldsymbol{0}_{3\times3}, \left[\hat{\boldsymbol{C}}_B^N\boldsymbol{a}^B\times\right]\left(\sigma_g^2\boldsymbol{I}_{3\times3} + \sum_{\mathrm{ALA}}\right)\left[\hat{\boldsymbol{C}}_B^N\boldsymbol{a}^B\times\right]^T, \boldsymbol{0}_{3\times3}\right)$$

and σ_{arw}^2 are the variance of the random walk noise of accelerometer biases.

The 9-DoF EKF's update equations presented in this article represent a novel approach that differs from the conventional method. Here, the measurement vector $\boldsymbol{z}_\beta$ is considered, which represents the difference between the measured distance vector $\boldsymbol{d}_m = [d_{1m}, d_{2m}, \ldots, d_{Nm}]^T$ and the actual distance vector $\boldsymbol{d} = [d_1, d_2, \ldots, d_N]^T$, where d is defined in (24), and N represents the number of base stations. The update equations can then be defined as follows:

$$\boldsymbol{z}_\beta = \boldsymbol{d}_m - \boldsymbol{d} \tag{32}$$

$$\boldsymbol{J} = \left[\partial\boldsymbol{d}/\partial x, \partial\boldsymbol{d}/\partial y, \partial\boldsymbol{d}/\partial z\right]_{N\times3} \approx \left[\boldsymbol{z}_\beta/\delta x, \boldsymbol{z}_\beta/\delta y, \boldsymbol{z}_\beta/\delta z\right]_{N\times3} \tag{33}$$

$$\boldsymbol{z}_\beta = \boldsymbol{J}\delta\boldsymbol{p} - 2\boldsymbol{n}_\beta = \boldsymbol{H}_\beta\delta\boldsymbol{\beta} + \boldsymbol{w}_\beta \tag{34}$$

$$\boldsymbol{H}_\beta = \left[\boldsymbol{J}, \boldsymbol{0}_{N\times6}\right]_{N\times9} \tag{35}$$

where $\boldsymbol{J}$ is the Jacobian matrix of the distance vector $\boldsymbol{d}$ and can be written in approximate format as shown in (33), where $[\delta x, \delta y, \delta z]^T$ represents the 3-D position error. In (34), the $\boldsymbol{w}_\beta$ term follows $N(0, \boldsymbol{R}_\beta)$, where $\boldsymbol{R}_\beta = ((1-\tau)\sigma_1^2 + \tau\sigma_2^2)\boldsymbol{I}_{N\times N}$, and $\boldsymbol{n}_\beta$ is the noise vector detailed in (24). The Jacobian matrix $\boldsymbol{J}$ can be derived by taking the partial derivative of the measured distance $d_{i,m}$, which yields $\partial d_{i,m}/\partial x_m = (x_m - x_i)/d_{i,m}$, $\partial d_{i,m}/\partial y_m = (y_m - y_i)/d_{i,m}$, and $\partial d_{i,m}/\partial z_m = (z_m - z_i)/d_{i,m}$.

Algorithm 1 Tightly Coupled EKF Framework

1. Initialize error state $\delta\boldsymbol{p}, \delta\boldsymbol{v}, \delta\boldsymbol{\psi}, \delta\boldsymbol{b}_a, \delta\boldsymbol{b}_g$.
2. Predict error state of each ALA-EKF
   $\delta\boldsymbol{\gamma}_{t|t-1} = \boldsymbol{\Phi}_{\gamma,t-1}\delta\boldsymbol{\gamma}_{t-1|t-1}$ (6-DoF EKF),
   $\delta\boldsymbol{\beta}_{t|t-1} = \boldsymbol{\Phi}_{\beta,t-1}\delta\boldsymbol{\beta}_{t-1|t-1}$ (9-DoF EKF).
3. Compute error covariance matrix
   $\boldsymbol{P}_{\gamma,t|t-1} = \boldsymbol{\Phi}_{\gamma,t-1}\boldsymbol{P}_{\gamma,t-1|t-1}\boldsymbol{\Phi}_{\gamma,t-1x}^T + \boldsymbol{Q}_{\gamma,t-1}$ (6-DoF EKF),
   $\boldsymbol{P}_{\beta,t|t-1} = \boldsymbol{\Phi}_{\beta,t-1}\boldsymbol{P}_{\beta,t-1|t-1}\boldsymbol{\Phi}_{\beta,t-1}^T + \boldsymbol{Q}_{\beta,t-1}$ (9-DoF EKF),
   where $\boldsymbol{Q}_\gamma = \sum_g + \sum_\gamma$ in (21), $\boldsymbol{Q}_\beta = \sum_a + \sum_\beta$ in (29).
4. Repeat steps 2-3 within the threshold of the predicted step.
5. Correct the Kalman gain
   $\boldsymbol{K}_{\gamma,t} = \boldsymbol{P}_{\gamma,t|t-1}\boldsymbol{H}_{\gamma,t}^T\left(\boldsymbol{H}_{\gamma,t}\boldsymbol{P}_{\gamma,t|t-1}\boldsymbol{H}_{\gamma,t}^T + \boldsymbol{R}_{\gamma,t}\right)^{-1}$ (6-DoF EKF),
   $\boldsymbol{K}_{\beta,t} = \boldsymbol{P}_{\beta,t|t-1}\boldsymbol{H}_{\beta,t}^T\left(\boldsymbol{H}_{\beta,t}\boldsymbol{P}_{\beta,t|t-1}\boldsymbol{H}_{\beta,t}^T + \boldsymbol{R}_{\beta,t}\right)^{-1}$ (9-DoF EKF),
   where $\boldsymbol{R}_\gamma = \sum_1 + \sum_2$ in (23).
6. Update the error covariance matrix
   $\boldsymbol{P}_{\gamma,t|t} = \left(\boldsymbol{I}_{6\times6} - \boldsymbol{K}_{\gamma,t}\boldsymbol{H}_{\gamma,t}\right)\boldsymbol{P}_{\gamma,t|t-1}$ (6-DoF EKF),
   $\boldsymbol{P}_{\beta,t|t} = \left(\boldsymbol{I}_{9\times9} - \boldsymbol{K}_{\beta,t}\boldsymbol{H}_{\beta,t}\right)\boldsymbol{P}_{\beta,t|t-1}$ (9-DoF EKF).
7. Update the error state
   $\delta\boldsymbol{\gamma}_{t|t} = \delta\boldsymbol{\gamma}_{t|t-1} + \boldsymbol{K}_{\gamma,t}\left(\boldsymbol{z}_{\gamma,t} - \boldsymbol{z}_{\gamma,t|t-1}\right)$ (6-DoF EKF),
   $\delta\boldsymbol{\beta}_{t|t} = \delta\boldsymbol{\beta}_{t|t-1} + \boldsymbol{K}_{\beta,t}\left(\boldsymbol{z}_{\beta,t} - \boldsymbol{z}_{\beta,t|t-1}\right)$ (9-DoF EKF).
8. Update the accelerometer bias of 6-DoF EKF and the corrected DCM of 9-DoF EKF in step 2, respectively.
9. Compute $\sum_{ALA}$ (10), and update $\boldsymbol{Q}_\gamma, \boldsymbol{Q}_\beta$ and $\boldsymbol{R}_r$, which will be used in step 3 and step 5, respectively.
10. Reset the error states and repeat steps 2-9 to achieve convergence in the gyroscope and accelerometer biases.

D. TC-EKF Framework

In the proposed tightly coupled EKF (TC-EKF) framework, the two EKFs detailed above are closely combined, with the 6-DoF EKF operating in the first phase and the 9-DoF EKF in the second. The attitude estimates from the first phase are used in the second phase to correct the 3-D position, while the accelerometer bias measurements from the second phase are fed back to refine the first phase. Furthermore, the velocity estimates from the second phase are used to obtain the LA, thus providing a robust ALA-based error covariance matrix estimation approach that addresses the previously described uncertainty issues. The TC-EKF framework is summarized in Algorithm 1.

IV. Computational Complexity Analysis

This section presents a comparison between the computational complexity of existing methods and that of the proposed approach. The complexity of matrix multiplication between a $U \times W$ matrix and a $W \times P$ matrix is $O(UWP)$, and the complexity of inverting a $W \times W$ square matrix is $O(W^3)$,

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where $O(\cdot)$ represents big O notation [34]. Note that we disregard the complexity of matrix addition, subtraction, and transposition, as the computational loads of these operations are much smaller. Therefore, the complexity of the EKF would be

$$O\left(3W^3 + (3U+1)W^2 + 2U(U+1)W + U^3\right) \tag{36}$$

where W denotes the number of states and U represents the row length of matrix $\boldsymbol{H}$.

For comparison, the complexity of existing MVCE and CI-based methods and the proposed approach are calculated numerically, where the Q matrices of the MVCE and CI can be written as follows:

$$\sum_{\text{MVCE}} = \alpha^{-1}\left(\boldsymbol{R}_1 + \boldsymbol{R}_2 + (\boldsymbol{m}_1 - \boldsymbol{m}_2)(\boldsymbol{m}_1 - \boldsymbol{m}_2)^T/4\right) \tag{37}$$

$$\sum_{\text{CI}} = c\,\text{diag}(\boldsymbol{\mu})^2 + \boldsymbol{R}. \tag{38}$$

The MVCE in (37) applies two ellipsoids that are used to calculate the covariance both with and without accounting for the uncertainty error; therefore, this approach involves two ellipsoids with means of and $\boldsymbol{m}_1$ and $\boldsymbol{m}_2$ covariance values of $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$, while α^{-1} denotes the maximum distance as the semimajor axis of the ellipsoid [19]. Therefore, the complexity of $\sum_{\text{MVCE}}$ is given by $O(2W^2 + W^{2n})$. In the CI method in (38), c is a constant, $\boldsymbol{\mu}$ is the residual error, and $\boldsymbol{R}$ is the covariance matrix. We disregard the complexity of $\sum_{\text{CI}}$ since it has a low computational load. Note that four UWB base stations are used in this study, meaning that the variable U in (36) is set to 4. Therefore, the complexity of a conventional 15-DoF EKF is $O(13714)$, while that of the MVCE-TC-EKF is $O(5878 + 9^{2n})$. The TC-EKF, CI-TC-EKF, and ALA-TC-EKF approaches have the same complexity of $O(5716)$.

In conclusion, the complexity of the proposed TC-EKF framework is reduced by up to 58.32% compared to that of the 15-DoF EKF framework. As the sample size n in MVCE increases, the computational complexity increases rapidly, whereas the computational complexity of the CI-TC-EKF and ALA-TC-EKF approaches remain constant and are much lower than that of MVCE-TC-EKF. This substantial reduction in complexity makes the proposed TC-EKF framework highly efficient for use in lower frequency microprocessor applications.

V. Simulation Results

The simulation performance of the proposed TC-EKF framework and ALA-TC-EKF methods is provided in this section, along with a comparison of the results with those of the 15-DoF-EKF, MVCE, and CI-based methods. The $\sum_{\text{MVCE}}$ and $\sum_{\text{CI}}$ terms are applied in both phases of the TC-EKF framework, with the matrices of $\sum_{\text{MVCE}}$ and $\sum_{\text{CI}}$ calculated independently. The attitude and 3-D localization accuracy are investigated using the root mean square attitude error (RMSAE) and root mean square position error (RMSPE), which are defined as follows:

$$\sum_{i=1}^{n}\left(\left(\hat{\phi}_i - \phi_i\right)^2 + \left(\hat{\theta}_i - \theta_i\right)^2 + \left(\hat{\psi}_i - \psi_i\right)^2\right)^{1/2}\Big/ n \tag{39}$$

TABLE I
Initial Values and Simulation Settings

Parameters	Values
Accelerometer	
Bias	0.1 g
1σ noise	0.01 g
1σ bias stability	0.1/3600 g/s^2
Gyroscope	
Bias	0.5 deg/s
1σ noise	0.02 deg/s
1σ rate random walk	0.02/3600 deg/s
Magnetometer	
Magnetic strength in Taipei	$\left\|\boldsymbol{m}^N\right\| = 45.488\ uT$
Inclination angle	37.45 deg
Declination angle	-4.57 deg
Bias	0.01 $\boldsymbol{m}^N$ uT
1σ noise	0.01 $\boldsymbol{m}^N$ uT
Random walk	0.01 $\boldsymbol{m}^N$/3600 uT/s^2
UWB	
Bias	0 m in LOS, 0.5 m in NLOS
1σ noise	0.1 m
Number of UWB sensors	5
Simulation Settings	
Sampling rate	100 Hz
Number of Monte-Carlo	80000
Location region	25×25×25 m^3

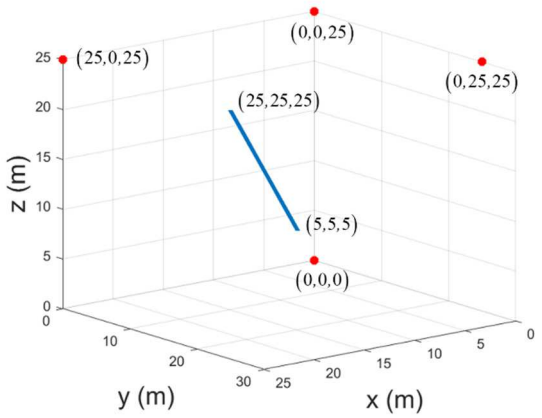


Fig. 2. Simulated inclined straight-line trajectory.

$$\sum_{i=1}^{n}\left((\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2 + (\hat{z}_i - z_i)^2\right)^{1/2}\Big/ n \tag{40}$$

where n is the sample size. The terms $[\hat{\phi}, \hat{\theta}, \hat{\psi}]^T$ and $[\hat{x}, \hat{y}, \hat{z}]^T$ represent the estimated attitude and location, while $[\phi, \theta, \psi]^T$ and $[x, y, z]^T$ represent the true attitude and location.

A. Simulation Scenario Settings

The simulation scenario settings and initial values are provided in Table I. Based on the optimal distribution of base

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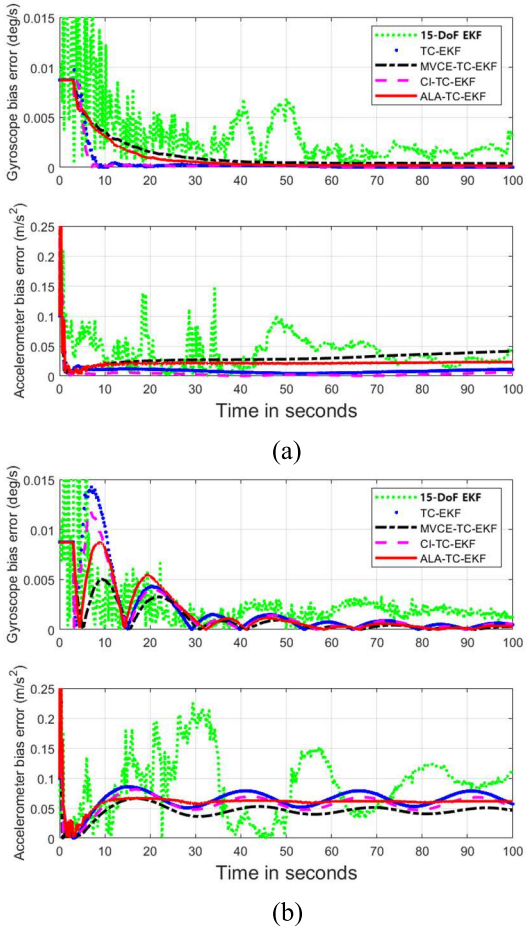


Fig. 3. Comparison of time responses of inertial sensor bias. (a) Maximum LA $\max(\|\boldsymbol{a}_l\|)$: 0.07 m/s^2, standard deviation of accelerometer noise (σ_a): 0.01 g, contamination ratio (τ): 10%, update rate ($1/\Delta t$): 100 Hz. (b) $\max(\|\boldsymbol{a}_l\|)$: 1.09 m/s^2, σ_a: 0.01 g, τ: 10%, and $1/\Delta t$: 100 Hz.

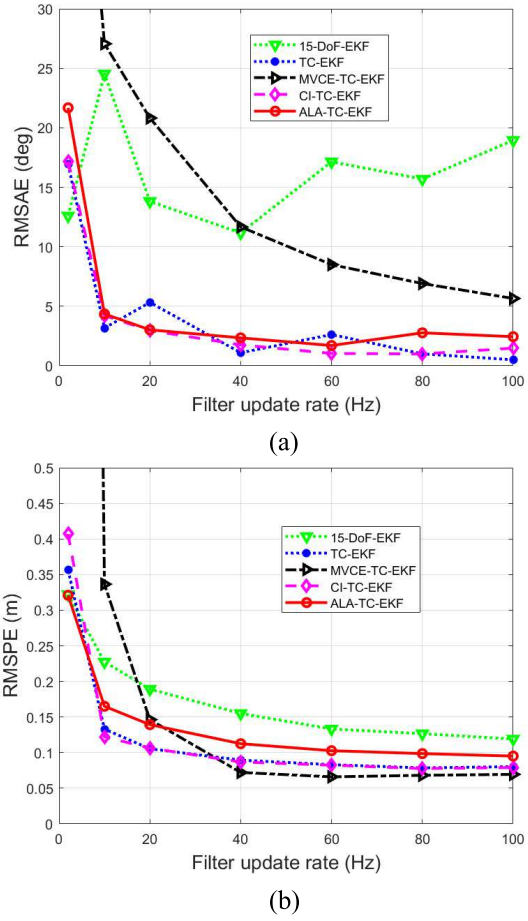


Fig. 4. RMSAE and RMSPE as a function of update rate. (a) $\max(\|\boldsymbol{a}_l\|)$: 0.07 m/s^2, σ_a: 0.01 g, and τ: 10%. (b) $\max(\|\boldsymbol{a}_l\|)$: 0.07 m/s^2, σ_a: 0.01 g, and τ: 10%.

stations in [35], the four simulated base stations are located at positions of (0, 0, 0), (0, 0, 25 m), (25, 0, 25 m), and (0, 25, 25 m). As shown in Fig. 2, the simulated trajectory is an inclined straight line, with the mobile node moving from the start point (5, 5, 5 m) to the endpoint (25, 25, 25 m). The simulation period is set to 200 s, with the roll, pitch, and yaw held constant at 0°, 0°, and 4.5°, respectively. The total simulation time is 800 s, generating 80 000 samples with a sampling frequency of 100 Hz. In addition, the mobile node undergoes reciprocal motion with variable acceleration along the simulated trajectory. Given the complexity of real indoor environments, all the simulations are conducted under mixed LOS/NLOS conditions.

B. Simulation Performance Comparison

The time responses of the IMU bias errors are shown in Fig. 3. As illustrated in Fig. 3(a), the maximum LA $\max(\|\boldsymbol{a}_l\|)$ is 0.07 m/s^2, with the standard deviation of the acceleration noise σ_a being 0.01 g, where g represents the gravitational constant. The simulated contamination ratio τ of the NLOS condition is 10%, and the EKF update rate is 100 Hz. As shown, the TC-EKF achieves faster convergence than the 15-DoF EKF, along with lower bias and noise error. In Fig. 3(b), the maximum LA is increased from 0.07 to 1.09 m/s^2, while the other conditions remain the same as previously. The convergence rate and bias error of the TC-EKF both perform better than the corresponding values of the 15-DoF EKF. In particular, the proposed ALA-TC-EKF exhibits greater stability under this scenario.

A brief sensitivity analysis is also performed to compare the proposed TC EKF and the 15-DoF EKF, both of which are based on a fixed-Q matrix, with noise standard deviations listed in Table I. As shown in Fig. 3(a), when the maximum LA is 0.07 m/s^2, the TC EKF exhibits much lower sensitivity. As the maximum LA increases to 1.09 m/s^2 [see Fig. 3(b)], sensitivity increases for both methods; however, the 15-DoF EKF becomes highly disturbed, while the TC EKF still maintains a low level of sensitivity. The proposed TC EKF provides less sensitivity to the sensor modeling errors (like state-dependent errors) either caused by hardware itself or unknown environments so that it can outperform the 15-DoF EKF.

Fig. 4 illustrates how the RMSAE and RMSPE vary with filter update rates, with the NLOS contamination ratio τ held constant at 10%. As shown, the accuracy of the TC-EKF-based algorithms improves significantly as the update rate increases. In Fig. 4(a), the RMSAE of the 15-DoF EKF does not converge due to the lack of attitude error estimation

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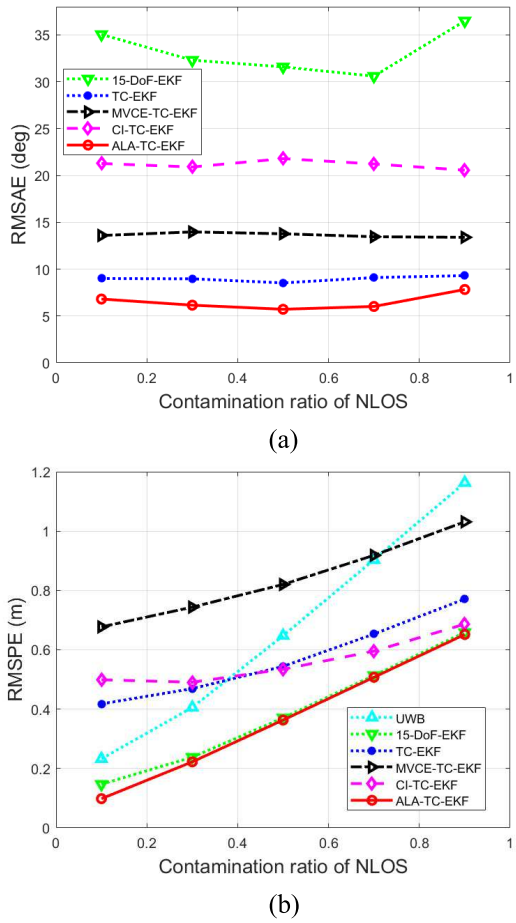


Fig. 5. RMSAE as a function of the contamination ratio of NLOS. (a) $\max(\|\boldsymbol{a}_l\|)$: 1.71 m/s^2, σ_a: 0.01 g, and $1/\Delta t$: 100 Hz. (b) $\max(\|\boldsymbol{a}_l\|)$: 1.71 m/s^2, σ_a: 0.01 g, and $1/\Delta t$: 100 Hz.

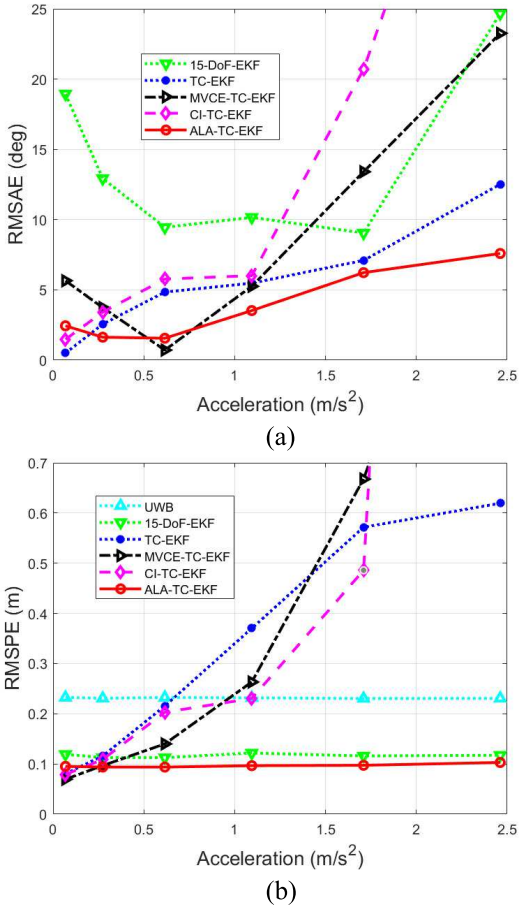


Fig. 6. RMSAE and RMSPE as a function of the maximum LA. (a) σ_a: 0.01 g, τ: 10%, and $1/\Delta t$: 100 Hz. (b) σ_a: 0.01 g, τ: 10%, and $1/\Delta t$: 100 Hz.

in its measurement stage [14], [32], [33]. The RMSAE of the TC-EKF-based methods approaches convergence when the update rate exceeds 20 Hz, whereas the MVCE-TC-EKF performs worse. The RMSAEs for the TC-EKF, CI-TC-EKF, and ALA-TC-EKF approaches are approximately 1.64°, while the minimum RMSAE of the MVCE-TC-EKF method is 5.65°. Both the TC-EKF and ALA-TC-EKF methods achieve the same RMSAE as the CI-TC-EKF, but their values are significantly lower than that of the MVCE-TC-EKF by 4.01°. As shown in Fig. 4(b), the RMSPEs of all methods converge after 20 Hz. The RMSPEs of the TC-EKF and ALA-TC-EKF approaches are approximately 0.09 and 0.11 m, respectively. The value for the CI-TC-EKF is nearly identical to that of the TC-EKF, while the MVCE-TC-EKF achieves a lower value of approximately 0.08 m. The RMSPE of the 15-DoF-EKF is the worst among these methods at 0.14 m. Therefore, when the filter update rate exceeds 20 Hz, the RMSPEs of the proposed TC-EKF and ALA-TC-EKF are equivalent to the CI-TC-EKF (around 0.03–0.05 m), which is lower than the 15-DoF-EKF, but higher than the MVCE-TC-EKF by 0.01–0.03 m.

Fig. 5 presents how the RMSAE and RMSPE change with the NLOS contamination ratio. As shown, the maximum LA $\max(\|\boldsymbol{a}_l\|)$ is 1.71 m/s^2, the standard deviation of acceleration noise (σ_a) is 0.01 g, and the EKF update rate is 100 Hz. The single UWB localization statistics, shown in aqua in Fig. 5(b), represent results from the least-squares approach using four range-based source locations [36]. The RMSAE of the ALA-TC-EKF is smaller than that of the other methods by 2.50°–14.60°. Notably, as shown in Fig. 5(b), if the NLOS contamination ratio τ is set to 0.1, indicating a relatively obstacle-free environment, the RMSPEs of the multisensor-based methods, i.e., MVCE-TC-EKF, CI-TC-EKF, and TC-EKF, are 0.2–0.45 m higher than that of the single UWB positioning. However, as the NLOS contamination ratio increases, the performance gap of the multisensor-based approaches gradually decreases, and these methods eventually surpass the performance of the single UWB. Notably, the proposed ALA-TC-EKF approach achieves the lowest RMSPE across the entire NLOS contamination ratio range from 0.1 to 0.9. In particular, the RMSPE of the ALA-TC-EKF method remains under 0.65 m even when the NLOS contamination ratio τ reaches 0.9.

C. Comparison of Performance in Various LAs

Fig. 6 shows how the RMSAE and RMSPE vary as the maximum LAs range from 0.07 to 2.46 m/s^2. As shown, the ALA-TC-EKF clearly outperforms existing methods. In Fig. 6(a), when the LA is below 1.1 m/s^2, the RMSAEs of the methods based on the TC-EKF remain under 6°.

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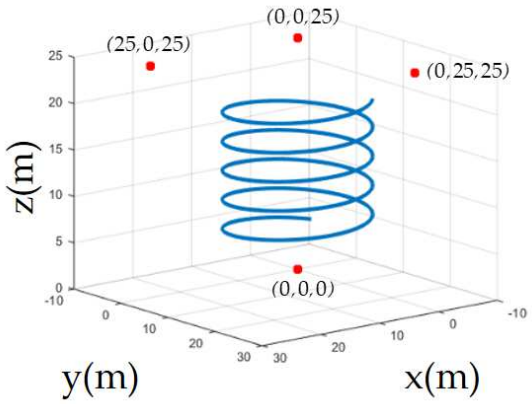


Fig. 7. Simulated helical trajectory.

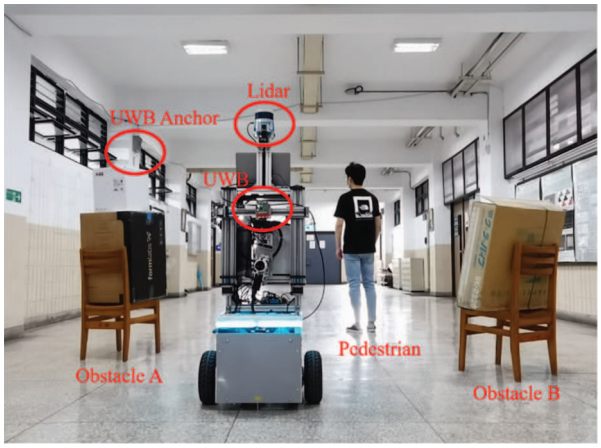
TABLE II
MEAN AND STD OF POSITION ERRORS

	15-DoF-EKF		CI-TC-EKF		ALA-TC-EKF	
$max(\|\boldsymbol{a}_l\|)$	Mean	Std	Mean	Std	Mean	Std
0.31	0.51	0.37	0.11	0.06	0.11	0.06
0.63	0.42	0.26	0.10	0.05	0.11	0.06
1.26	0.55	0.27	0.14	0.07	0.09	0.05
2.53	1.03	0.54	0.41	0.44	0.11	0.07
5.07	1.74	0.99	0.61	0.60	0.13	0.07
m/s^2	m					

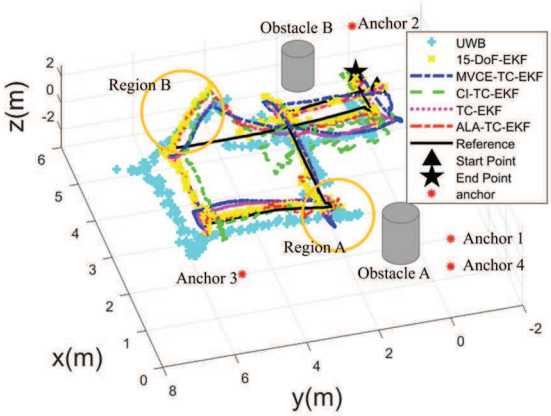
However, as the LA increases, the RMSAEs of the CI-TC-EKF, MVCE-TC-EKF, and TC-EKF increase rapidly, whereas the ALA-TC-EKF delivers superior performance, with the RMSAE remaining below 7.5° even when the LA reaches 2.5 m/s^2. Similarly, in Fig. 6(b), the RMSPEs of the other methods increase significantly as the LA increases, whereas those of the proposed ALA-TC-EKF remain stable. This demonstrates that the ALA-TC-EKF effectively mitigates the impact of LA on positioning error, achieving a stable accuracy within 0.1 m. Notably, the RMSPEs of the single UWB and 15-DoF-EKF also remain nearly constant, because the LA error is not accounted for in these two methods.

D. Comparison of Performance in Highly Agile Motions

To evaluate scenarios where the attitude changes over time, this section introduces a helical trajectory motion with the target moving along a helical path centered around a vertical line, as shown in Fig. 7, while its roll, pitch, and yaw angles vary continuously over time. This scenario applies three approaches, including the 15-DoF-, CI-TC-, and ALA-TC-based EKF, evaluating five cases with different maximum LAs, ranging from 0.31 to 5.07 m/s^2. The mean and standard (Std) deviations of the position errors are shown in Table II. From the data statistics, both the CI-TC-EKF and ALA-TC-EKF methods demonstrate significantly better accuracy and stability compared to the standard 15-DoF-EKF. The results indicate that the 15-DoF-EKF exhibits poor accuracy under dynamic motion with persistent angular variation. As LA increases,



(a)



(b)

Fig. 8. Laboratory experimental of rectangular trajectory. (a) Laboratory experimental environment. (b) Localization performance of the rectangular motion.

the proposed ALA-TC-EKF method achieves the best performance. When the maximum acceleration reaches 5.07 m/s^2, the ALA-TC-EKF improves accuracy by approximately 80% compared with CI-TC-EKF and by 93% compared with the standard 15-DoF-EKF.

VI. EXPERIMENTAL RESULTS

The studied methods were also tested in a laboratory environment. As shown in Fig. 8(a), four DecaWave DWM1000 UWB base stations were installed on the walls, and the mobile node was mounted on a LiDAR-equipped robot. The robot moved along a rectangular trajectory with varying acceleration. Several obstacles were placed within the testing environment, including obstacle A (a carton) located at coordinates (1.6, 1.8 m) and obstacle B at coordinates (3.3, 1.8 m). In addition, a tester walked back and forth along a straight line. Obstacle A blocked base stations No. 1 and No. 4, while obstacle B blocked base station No. 2. In the implementation, the IMU operates at 1000 Hz, while the MAG and UWB operate at 100 Hz. To synchronize the sensor data, the system runs at 100 Hz. Consequently, the experimental data was recorded at 100 Hz for a duration of 80 s, resulting in a total of 8000 samples.

After collecting the data, the samples were imported into a computer, and the positioning performance of the

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TABLE III
STATISTICS OF LOCALIZATION ERROR

	Simulation		Lab experiment	
	angle error	position error	RMSAE	RMSPE
UWB	--	0.43	--	1.68
15-DoF EKF	20.28	0.22	34.48	0.55
TC EKF	6.03	0.32	2.80	0.64
MVCE TC EKF	14.62	0.69	2.34	0.69
CI-TC EKF	12.27	0.56	4.53	0.75
ALA-TC EKF	5.21	0.19	1.96	0.57
Unit	*deg*	*m*	*deg*	*m*

15-DoF-EKF, TC-EKF, MVCE-TC-EKF, CI-TC-EKF, and ALA-TC-EKF algorithms was verified using MATLAB. As shown in Fig. 8(b), the results show that the proposed ALA-TC-EKF and 15-DoF-EKF methods achieve the trajectory closest to the reference, followed by the TC-EKF approach; then, the MVCE-TC-EKF approach, and finally, the CI-TC-EKF method, which performed the worst. Notably, the signal blocking caused by obstacle A resulted in more outliers in regions A and B due to the NLOS radio propagation. This is because the ideal knowledge of the NLOS ratio in (24) is used to simulate the measurement degradation under NLOS, which may not hold in real-world applications. In this scenario, UWB measurements exhibit severe outliers and highly non-Gaussian noise characteristics. Accurately modeling such conditions remains a significant challenge. The RMSAE and RMSPE statistics of the existing methods, along with the average values across different simulations of the inclined straight-line motion, are summarized in Table III. As shown, although the 15-DoF EKF achieves good positioning accuracy, it fails to correct angular errors. Overall, the proposed ALA-TC-EKF approach achieves near-optimal accuracy in both attitude and positioning estimation across all scenarios.

VII. CONCLUSION

This article aimed to modify the Q and R matrices from (22), (23), and (31) forming part of the proposed 6-DoF EKF and 9-DoF EKF designs to achieve a near-optimal solution similar to that obtained from a 15-DoF EKF. The proposed approach offers a 58.32% reduction in computational complexity. In addition, the proposed ALA-TC-EKF achieves fast convergence when the update rate exceeds 20 Hz, with the RMSAE limited to a maximum of 1.64° and the RMSPE limited to 0.11 m. As the NLOS contamination ratio increases, the ALA-TC-EKF consistently achieves the lowest attitude and position errors among the studied approaches. Notably, when the NLOS contamination ratio reaches 90%, the ALA-TC-EKF still maintains RMSAE and RMSPE values within 7.5° and 0.65 m, respectively. In addition, while an increase in LA to 2.5 m/s^2 would typically lead to divergence, the ALA-TC-EKF remains stable and its RMSAE and RMSPE are limited within 7.5° and 0.1 m, respectively. In conclusion, the proposed ALA-TC-EKF outperforms the other methods under conditions of severe LA, effectively mitigating the impact of translation motion and ensuring stable performance. Extensive simulation and experimental results demonstrate that the ALA-TC-EKF can maintain excellent stability even under large LAs. Although the proposed method achieves great improvements in terms of performance and efficiency relative to existing methods, a significant reduction in location accuracy is still observed under severe NLOS conditions due to outliers. In the future, a learning-based outlier classification method will be developed, utilizing machine learning-based statistics to eliminate outliers and improve the accuracy of these techniques.

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